



## **Robotics Lab Report**

### **Lab 5: Forward Kinematics**

**Name:** Demarce Williams

**Course:** Robotics

**Instructor:** Alexander Avdeev

**Semester:** Fall 2025

## **1. Introduction**

Forward kinematics (FK) determines the end-effector pose (position and orientation) given the robot's joint parameters.

This lab explores FK for three types of manipulators using both analytical and simulation approaches:

- 3-Link Planar Arm
- Spherical Arm
- Anthropomorphic Arm

### **Objective:**

- Derive the Denavit–Hartenberg (D-H) parameters for each manipulator.
- Compute transformation matrices symbolically and numerically.
- Simulate the manipulators in MATLAB and Simscape Multibody.
- Compare the analytical and simulated end-effector positions.

## 2. Manipulator 1 – 3-Link Planar Arm

Parameters:

| Link  | Length (cm) |
|-------|-------------|
| $l_1$ | 40          |
| $l_2$ | 20          |
| $l_3$ | 10          |

### 2.1 Coordinate Frame Assignment

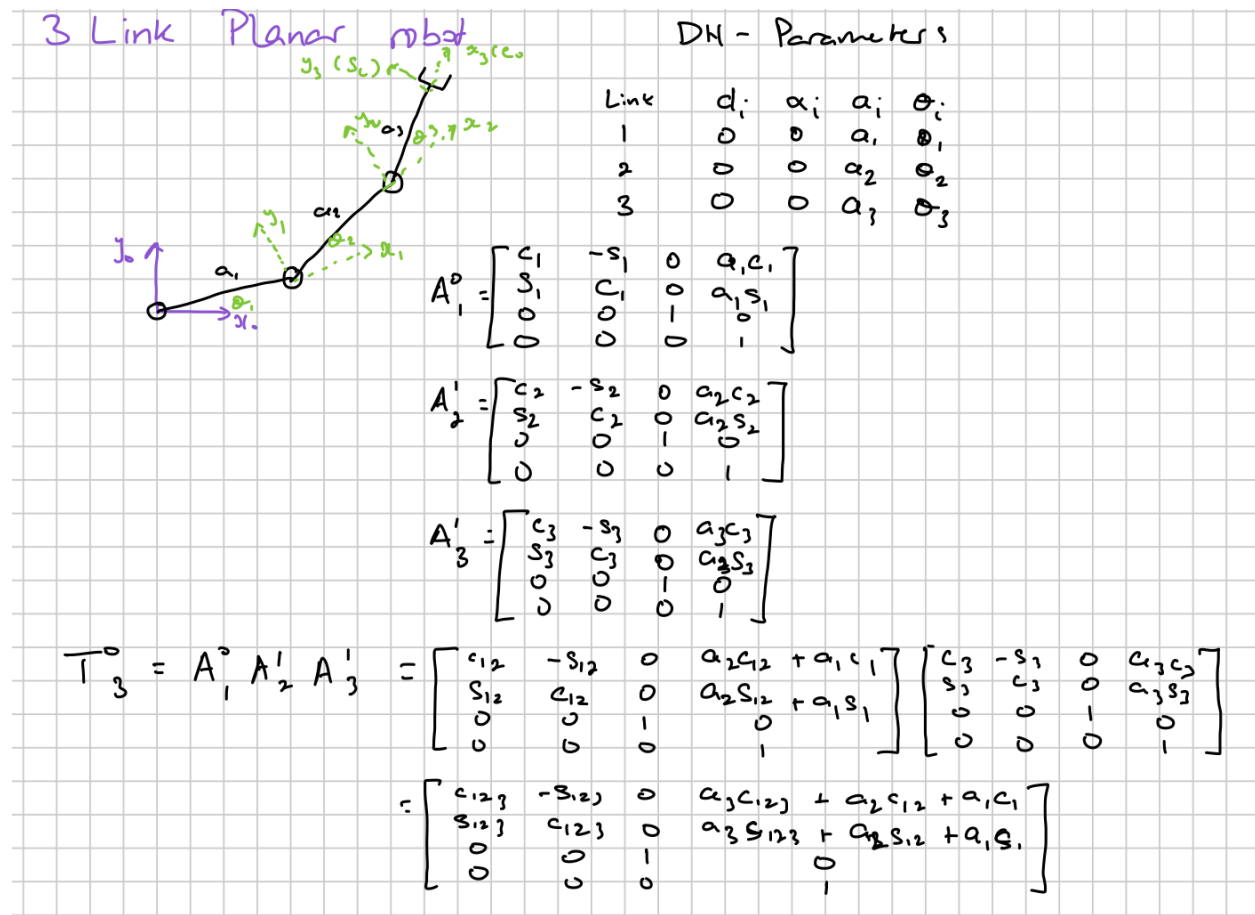


Figure 1: 3-Link Planar Arm Coordinate Frames and Hand Derivations

## 2.2 D-H Parameters

| Link | $a_i$ (m) | $\alpha_i$ (rad) | $d_i$ (m) | $\theta_i$ (rad) |
|------|-----------|------------------|-----------|------------------|
| 1    | a1        | 0                | 0         | $\theta_1$       |
| 2    | a2        | 0                | 0         | $\theta_2$       |
| 3    | a3        | 0                | 0         | $\theta_3$       |

## 2.3 Symbolic Derivation

### General Transformation Matrix Definition

--- General Transformation Matrix A ---

A =

$$\begin{pmatrix} \cos(\theta) & -\cos(\alpha) \sin(\theta) & \sin(\alpha) \sin(\theta) & a \cos(\theta) \\ \sin(\theta) & \cos(\alpha) \cos(\theta) & -\sin(\alpha) \cos(\theta) & a \sin(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) & d \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

### 3 Link Planar Robot

=== 1. 3-Link Planar Robot ===

A1 (Link 1 to Base):

$$\begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & a_1 \cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & a_1 \sin(\theta_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A2 (Link 2 to Link 1):

$$\begin{pmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & a_2 \cos(\theta_2) \\ \sin(\theta_2) & \cos(\theta_2) & 0 & a_2 \sin(\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A3 (Link 3 to Link 2):

$$\begin{pmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & a_3 \cos(\theta_3) \\ \sin(\theta_3) & \cos(\theta_3) & 0 & a_3 \sin(\theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T (Base to End Effector):

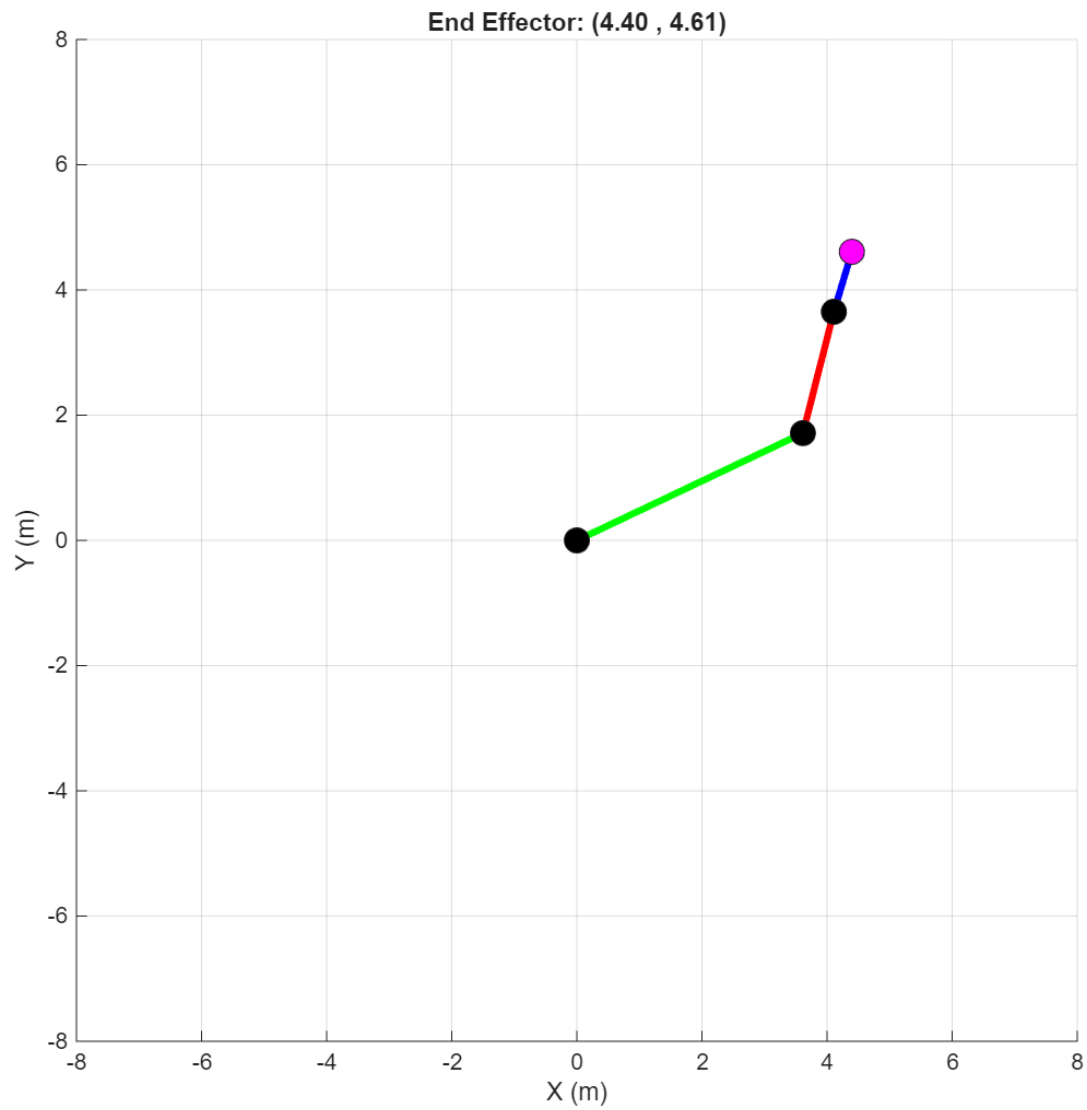
$$\begin{pmatrix} \sigma_2 & -\sigma_1 & 0 & a_2 \cos(\theta_1 + \theta_2) + a_1 \cos(\theta_1) + a_3 \sigma_2 \\ \sigma_1 & \sigma_2 & 0 & a_2 \sin(\theta_1 + \theta_2) + a_1 \sin(\theta_1) + a_3 \sigma_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where

$$\sigma_1 = \sin(\theta_1 + \theta_2 + \theta_3)$$

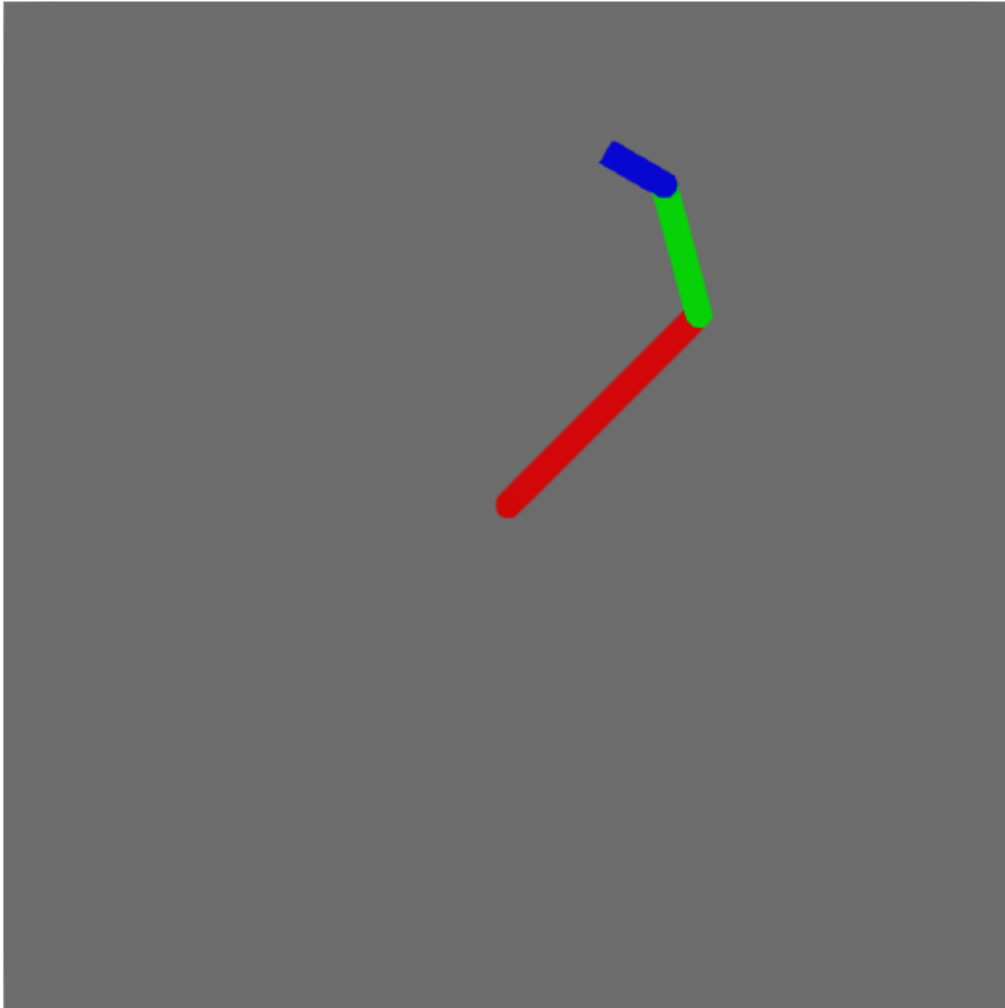
$$\sigma_2 = \cos(\theta_1 + \theta_2 + \theta_3)$$

## 2.4 Matlab Model

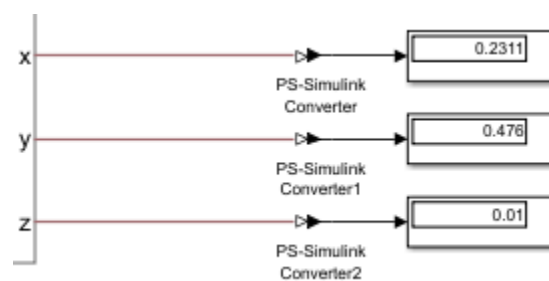


**Figure 2:** Matlab Model of the 3 Link Planar Arm

## 2.5 Simscape Model

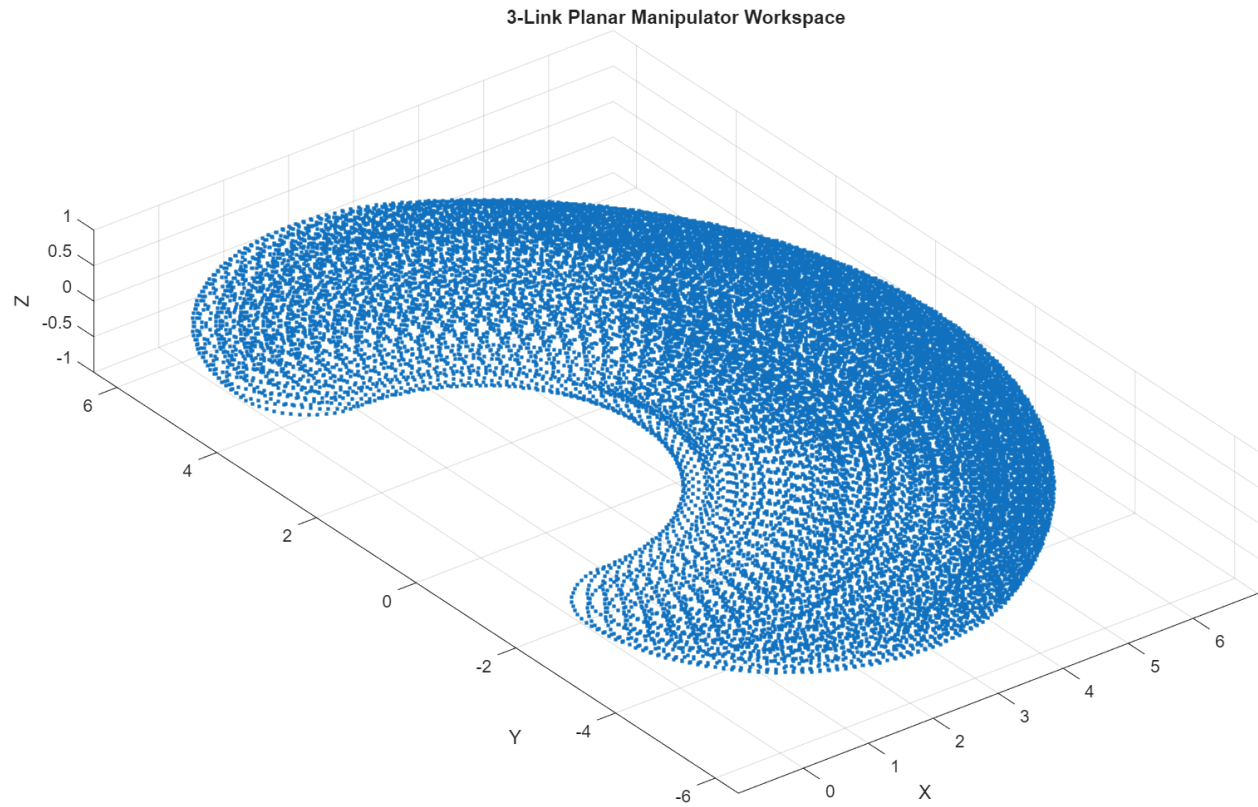


**Figure 3:** Simscape Model of the 3 Link Planar Arm



**Figure 4:** End-Effector XYZ Display in Simscape

## 2.6 Workspace Plot



*The workspace was plotted in cm with Links lengths scaled down by a factor of 10 so as to fit it into the grid. Also,  $\theta_1$  was constrained to  $[-\pi/3, \pi/3]$ ,  $\theta_2$  to  $[-2\pi/3, 2\pi/3]$ , and  $\theta_3$  to  $[-\pi/2, \pi/2]$ .*

**Figure 5:** Workspace of the 3-Link Planar Arm



### 3. Manipulator 2 – Spherical Arm

Parameters:

| Link  | Description     | Range / Length |
|-------|-----------------|----------------|
| $l_1$ | Base height     | 50 cm          |
| $l_2$ | Arm length      | 25 cm          |
| $d_3$ | Prismatic range | 20–80 cm       |

#### 3.1 Coordinate Frame Assignment

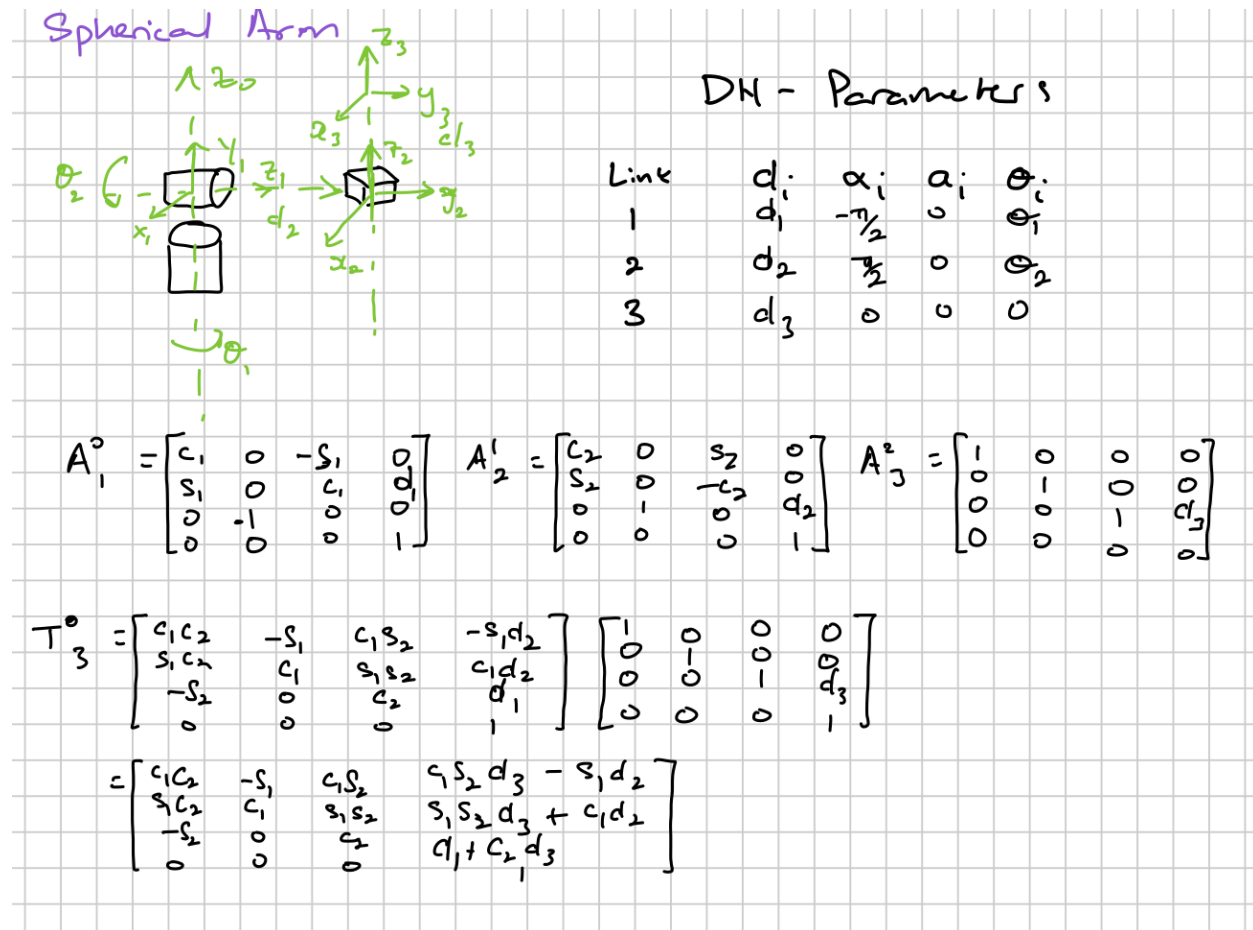


Figure 6: Spherical Arm Coordinate Frames and Hand Derivations

### 3.2 D-H Parameters

| Link | $a_i$ (m) | $\alpha_i$ (rad) | $d_i$ (m) | $\theta_i$ (rad) |
|------|-----------|------------------|-----------|------------------|
| 1    | 0         | $-\pi/2$         | $d_1$     | $\theta_1$       |
| 2    | 0         | $\pi/2$          | $d_2$     | $\theta_2$       |
| 3    | 0         | 0                | $d_3$     | 0                |

### 3.3 Symbolic Derivation

## General Transformation Matrix Definition

--- General Transformation Matrix A ---

A =

$$\begin{pmatrix} \cos(\theta) & -\cos(\alpha) \sin(\theta) & \sin(\alpha) \sin(\theta) & a \cos(\theta) \\ \sin(\theta) & \cos(\alpha) \cos(\theta) & -\sin(\alpha) \cos(\theta) & a \sin(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) & d \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

## Spherical Arm

=== 2. Spherical Arm ===

A1 (Link 1 to Base):

$$\begin{pmatrix} \cos(\theta_1) & 0 & -\sin(\theta_1) & 0 \\ \sin(\theta_1) & 0 & \cos(\theta_1) & 0 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A2 (Link 2 to Link 1):

$$\begin{pmatrix} \cos(\theta_2) & 0 & \sin(\theta_2) & 0 \\ \sin(\theta_2) & 0 & -\cos(\theta_2) & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

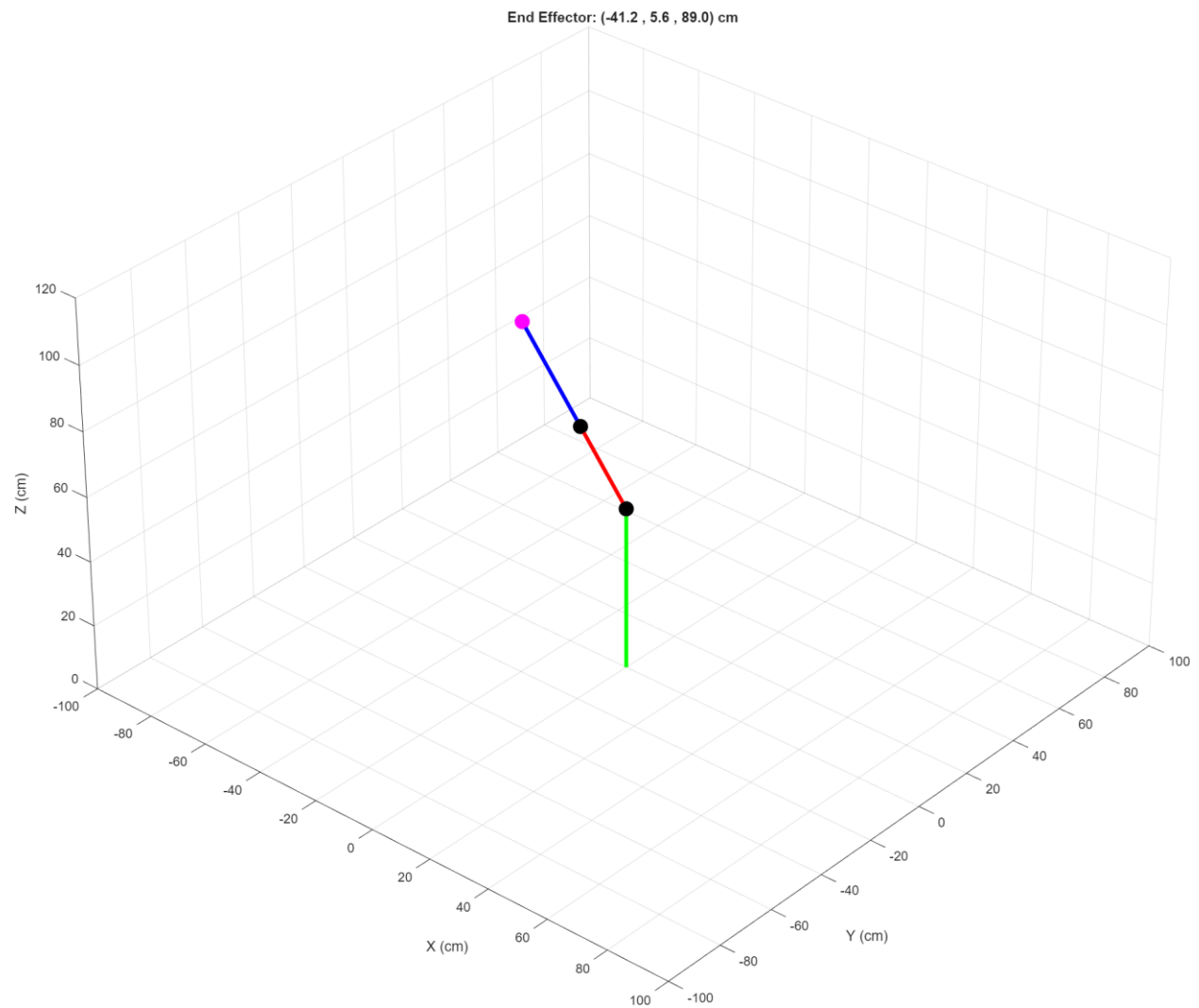
A3 (Link 3 to Link 2):

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T (Base to End Effector):

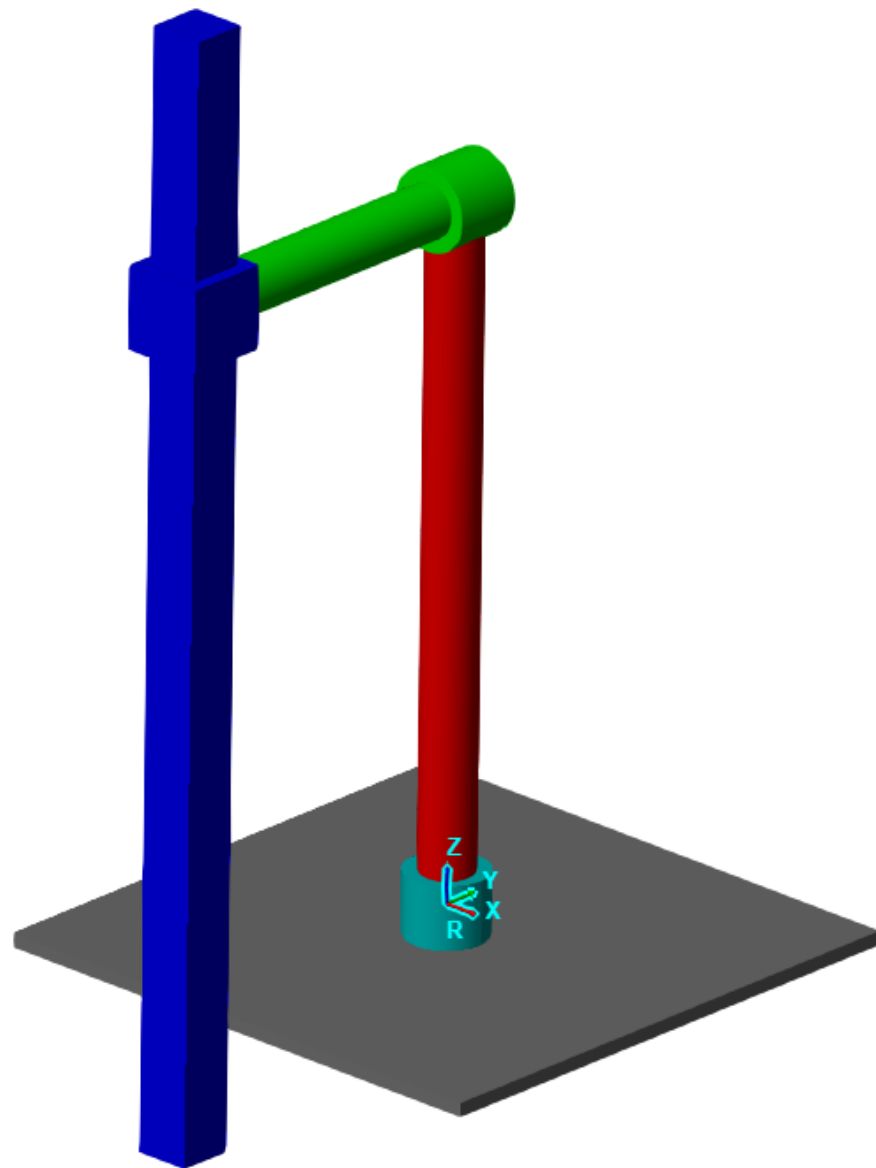
$$\begin{pmatrix} \cos(\theta_1) \cos(\theta_2) & -\sin(\theta_1) & \cos(\theta_1) \sin(\theta_2) & d_3 \cos(\theta_1) \sin(\theta_2) - d_2 \sin(\theta_1) \\ \cos(\theta_2) \sin(\theta_1) & \cos(\theta_1) & \sin(\theta_1) \sin(\theta_2) & d_2 \cos(\theta_1) + d_3 \sin(\theta_1) \sin(\theta_2) \\ -\sin(\theta_2) & 0 & \cos(\theta_2) & d_1 + d_3 \cos(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

### 3.4 Matlab Model

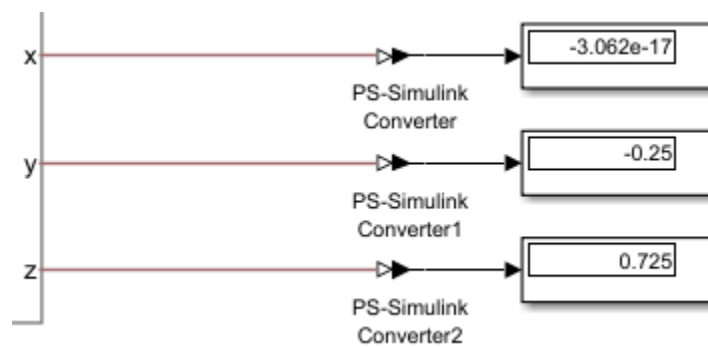


**Figure 7:** Matlab Model of the Spherical Arm

### 3.5 Simscape Model

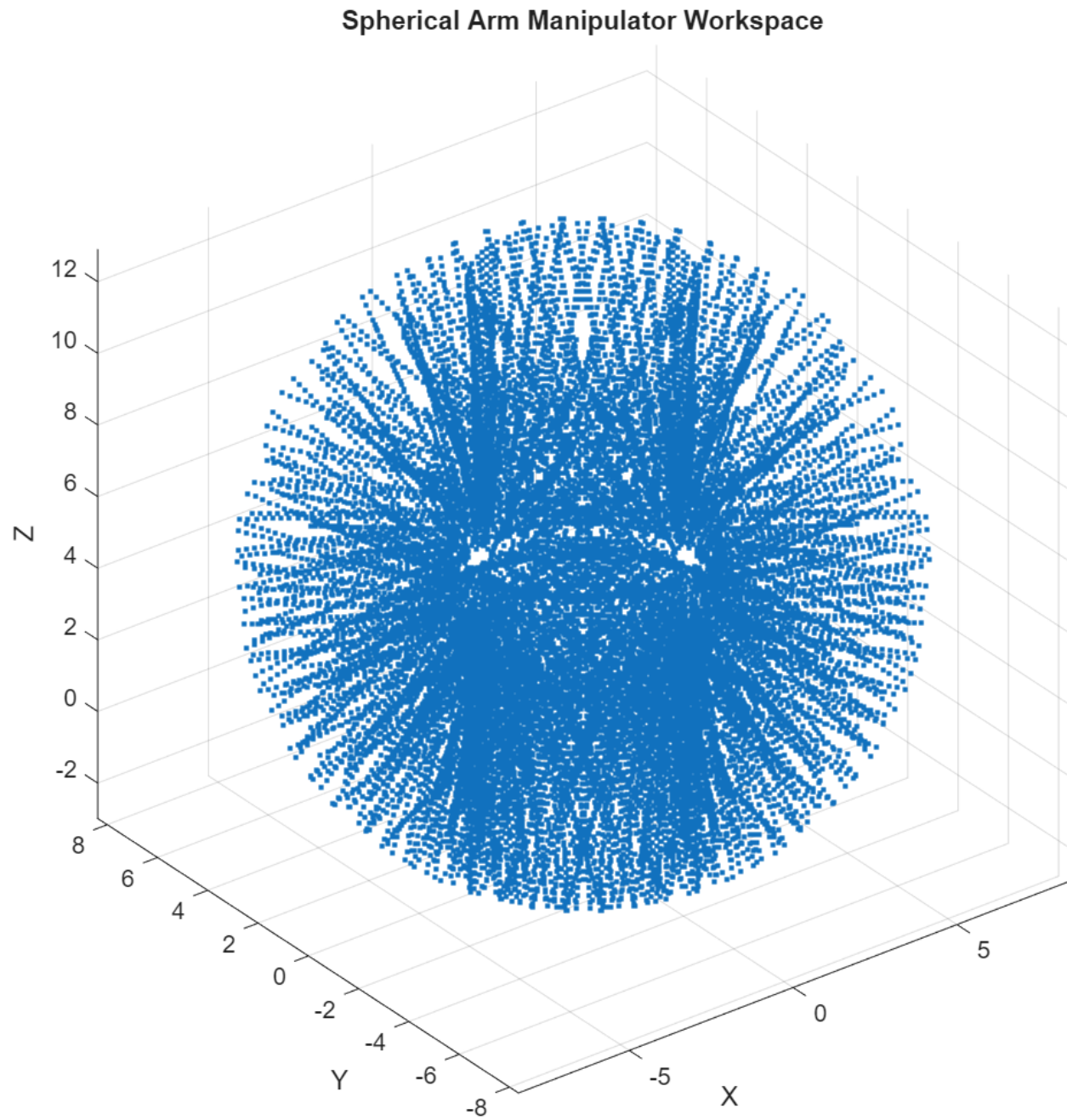


**Figure 8:** Simscape Model of the Spherical Arm



**Figure 9:** End-Effector XYZ Display in Simscape

### 3.5 Workspace Plot



*The workspace was plotted in cm with Links lengths scaled down by a factor of 10 so as to fit it into the grid. Also,  $\theta_1$  was constrained to  $[0, 2\pi]$ ,  $\theta_2$  to  $[-3\pi/2, 3\pi/2]$ , and  $d_3$  to  $[20, 80]$  cm.*

**Figure 10:** Workspace of the Spherical Arm

#### 4. Manipulator 3 – Anthropomorphic Arm

Parameters:

| Link  | Length (cm) |
|-------|-------------|
| $l_1$ | 70          |
| $l_2$ | 50          |
| $l_3$ | 50          |

##### 4.1 Coordinate Frame Assignment

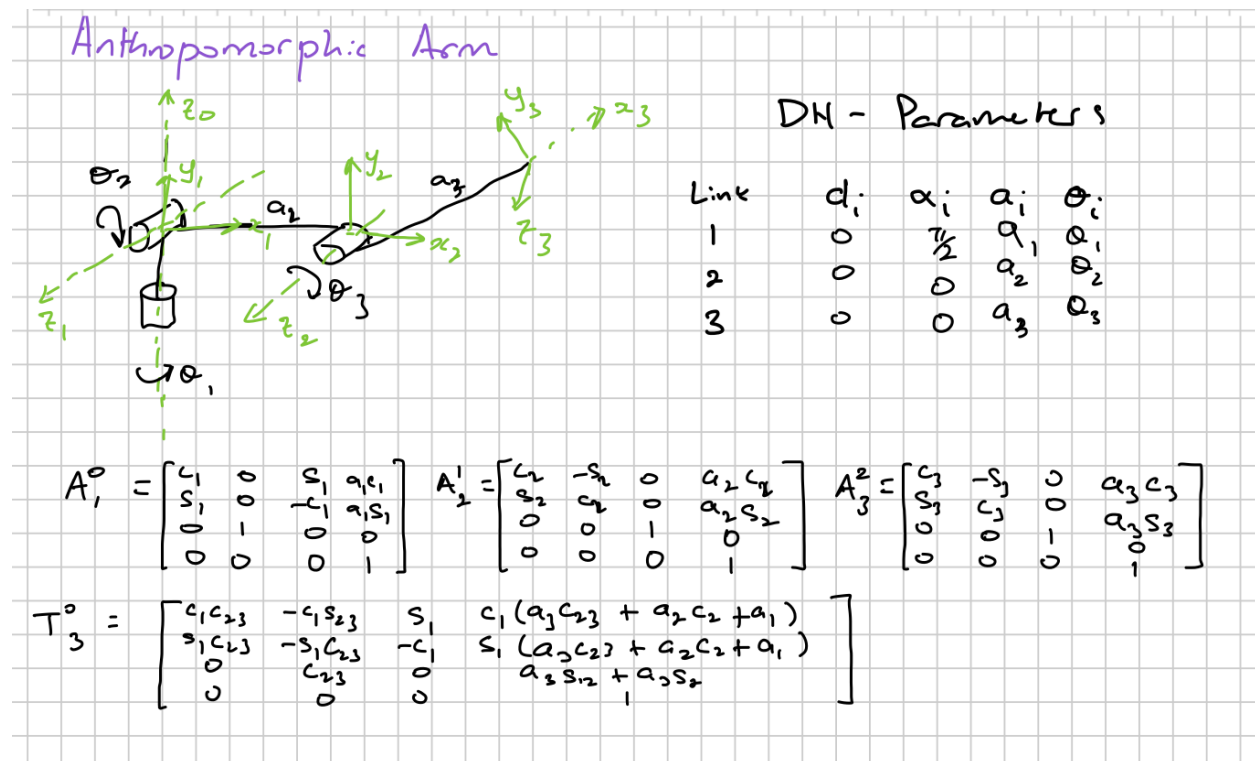


Figure 11: Anthropomorphic Arm Coordinate Frames and Hand Derivations

## 4.2 D-H Parameters

| Link | $a_i$ (m) | $\alpha_i$ (rad) | $d_i$ (m) | $\theta_i$ (rad) |
|------|-----------|------------------|-----------|------------------|
| 1    | $a_1$     | $\pi/2$          | 0         | $\theta_1$       |
| 2    | $a_2$     | 0                | 0         | $\theta_2$       |
| 3    | $a_3$     | 0                | 0         | $\theta_3$       |

## 4.3 Symbolic Derivation

### General Transformation Matrix Definition

--- General Transformation Matrix A ---

A =

$$\begin{pmatrix} \cos(\theta) & -\cos(\alpha) \sin(\theta) & \sin(\alpha) \sin(\theta) & a \cos(\theta) \\ \sin(\theta) & \cos(\alpha) \cos(\theta) & -\sin(\alpha) \cos(\theta) & a \sin(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) & d \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



## Anthropomorphic Arm

=== 3. Anthropomorphic Arm ===

A1 (Link 1 to Base):

$$\begin{pmatrix} \cos(\theta_1) & 0 & \sin(\theta_1) & a_1 \cos(\theta_1) \\ \sin(\theta_1) & 0 & -\cos(\theta_1) & a_1 \sin(\theta_1) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A2 (Link 2 to Link 1):

$$\begin{pmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & a_2 \cos(\theta_2) \\ \sin(\theta_2) & \cos(\theta_2) & 0 & a_2 \sin(\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A3 (Link 3 to Link 2):

$$\begin{pmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & a_3 \cos(\theta_3) \\ \sin(\theta_3) & \cos(\theta_3) & 0 & a_3 \sin(\theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

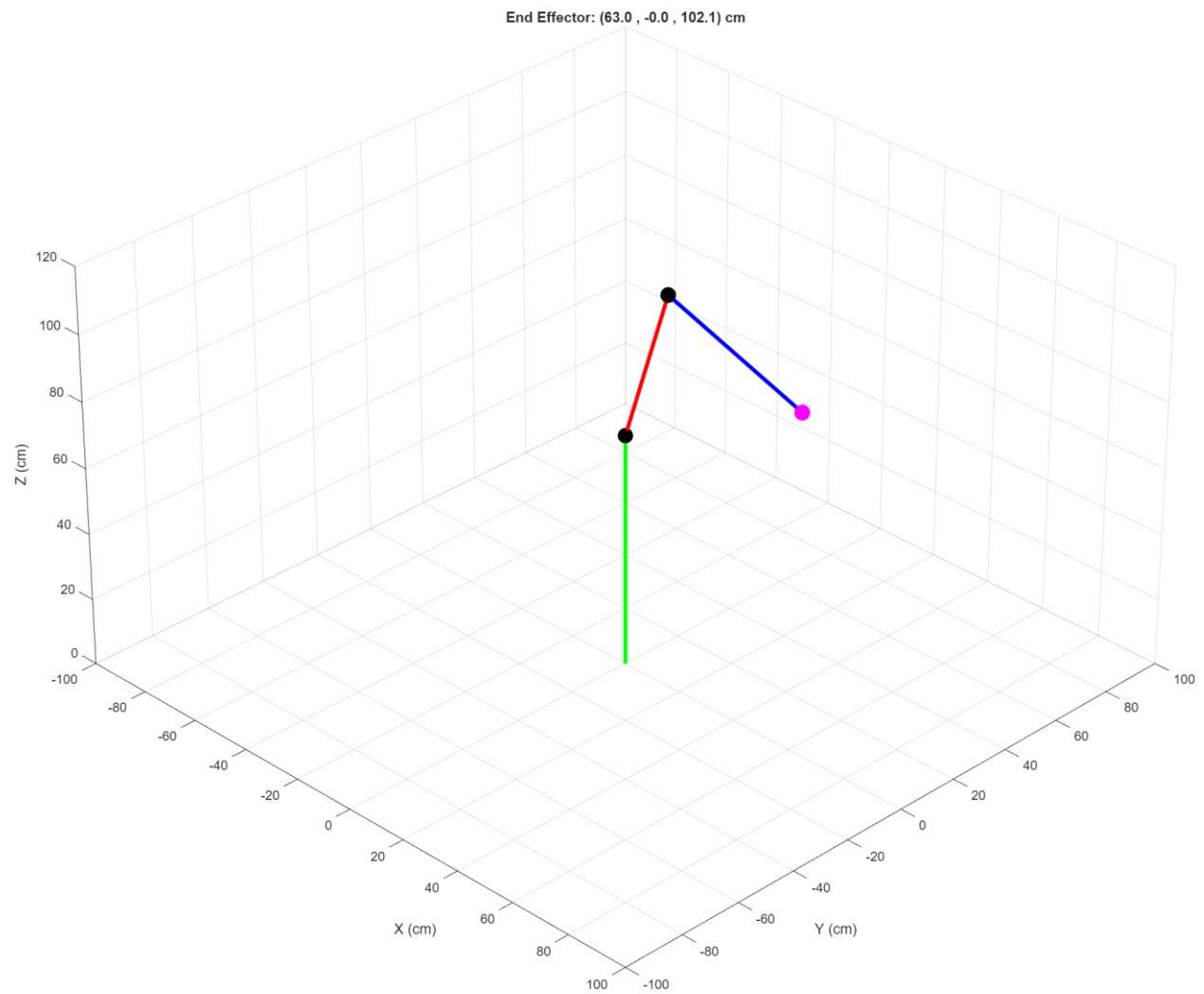
T (Base to End Effector):

$$\begin{pmatrix} \cos(\theta_2 + \theta_3) \cos(\theta_1) & -\sin(\theta_2 + \theta_3) \cos(\theta_1) & \sin(\theta_1) & \cos(\theta_1) \sigma_1 \\ \cos(\theta_2 + \theta_3) \sin(\theta_1) & -\sin(\theta_2 + \theta_3) \sin(\theta_1) & -\cos(\theta_1) & \sin(\theta_1) \sigma_1 \\ \sin(\theta_2 + \theta_3) & \cos(\theta_2 + \theta_3) & 0 & a_3 \sin(\theta_2 + \theta_3) + a_2 \sin(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where

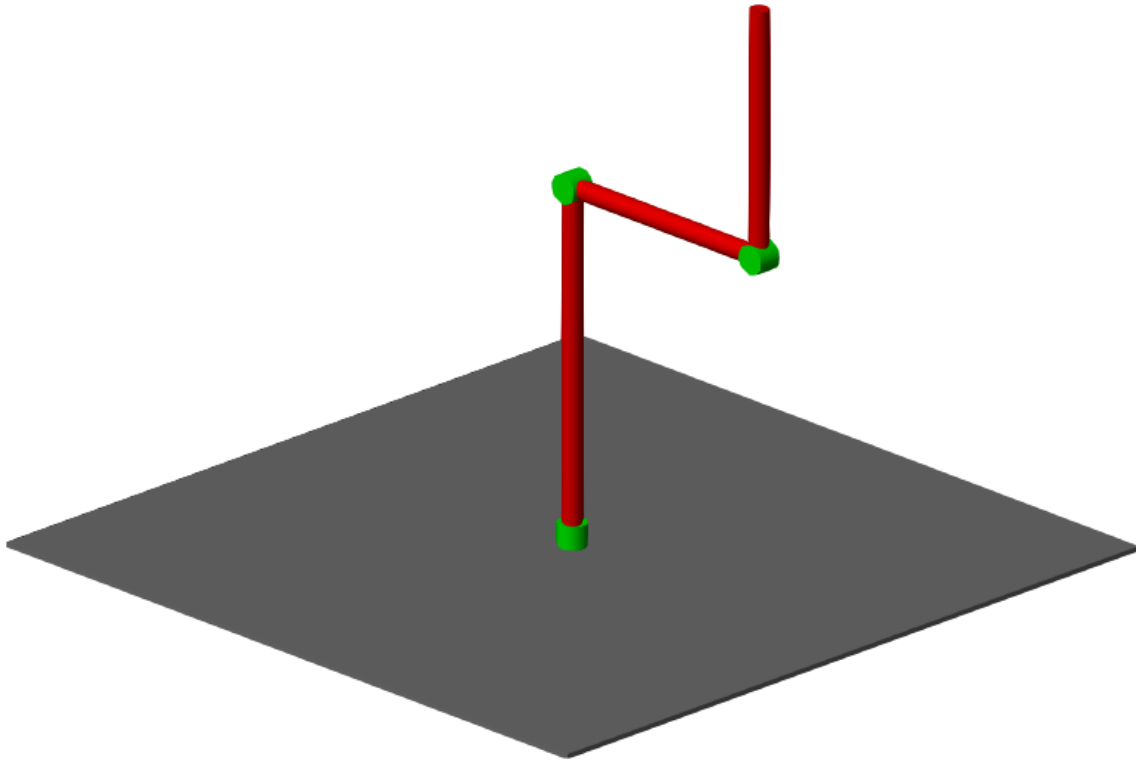
$$\sigma_1 = a_1 + a_3 \cos(\theta_2 + \theta_3) + a_2 \cos(\theta_2)$$

## 4.4 Matlab Model

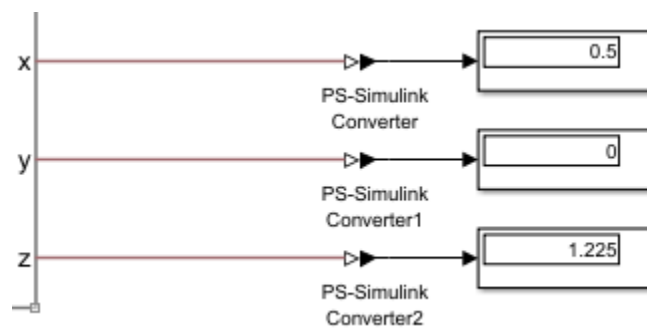


**Figure 12:** Matlab Model of the Anthrop. Arm

## 4.5 Simscape Model

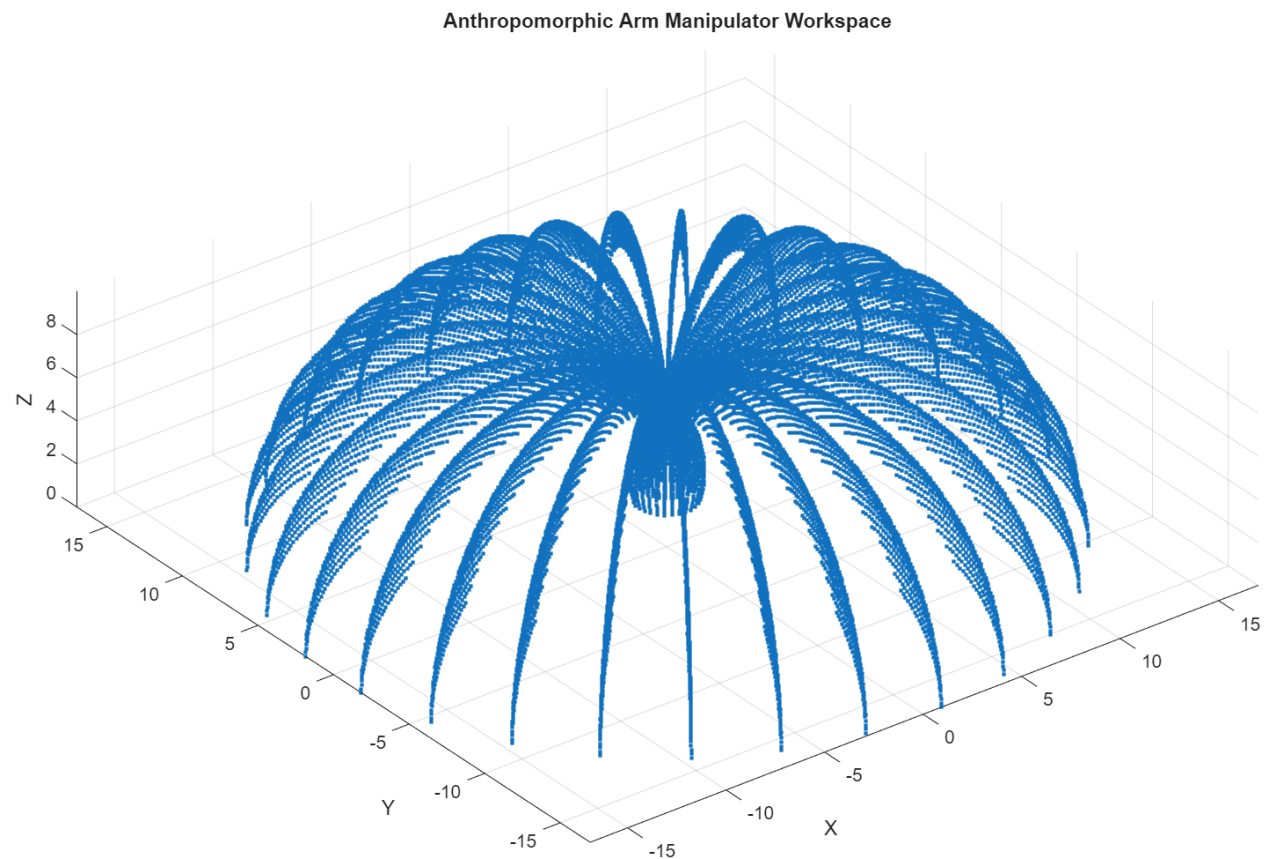


**Figure 13:** Simscape Model of the Anthrop. Arm



**Figure 14:** End-Effector XYZ Display in Simscape

## 4.5 Workspace Plot



*The workspace was plotted in cm with Links lengths scaled down by a factor of 10 so as to fit it into the grid. Also,  $\theta_1$  was constrained to  $[0, 2\pi]$ ,  $\theta_2$  to  $[0, 3\pi/4]$ , and  $\theta_3$  to  $[0, \pi/3]$ .*

**Figure 15:** Workspace of the Anthropol. Arm

## 5. Spherical Wrist with Gripper

**Task:** Create a Simscape model of a spherical wrist equipped with a prismatic joint gripper.

Spherical Wrist With Gripper

DH - Parameters

| Link | $d_i$ | $\alpha_i$       | $a_i$ | $\theta_i$ |
|------|-------|------------------|-------|------------|
| 1    | 0     | $-\frac{\pi}{2}$ | 0     | $\theta_1$ |
| 2    | 0     | $\frac{\pi}{2}$  | 0     | $\theta_2$ |
| 3    | $d_3$ | 0                | 0     | $\theta_3$ |

$$A_1^0 = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_2^1 = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_3^2 = \begin{bmatrix} c_3 & -s_3 & 0 & 0 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = \begin{bmatrix} c_1 c_2 c_3 - s_1 s_3 & -c_1 c_2 s_3 - s_1 c_3 & c_1 s_2 & c_1 s_2 d_3 \\ s_1 c_2 c_3 - c_1 s_3 & s_1 c_2 s_3 + c_1 c_3 & s_1 s_2 & s_1 s_2 d_3 \\ -s_2 c_3 & s_2 s_3 & c_2 & c_2 d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

## 5.1 DH Parameters

| Link | $a_i$ (m) | $\alpha_i$ (rad) | $d_i$ (m) | $\theta_i$ (rad) |
|------|-----------|------------------|-----------|------------------|
| 1    | 0         | $-\pi/2$         | 0         | $\theta_1$       |
| 2    | 0         | $\pi/2$          | 0         | $\theta_2$       |
| 3    | 0         | 0                | $d_3$     | $\theta_3$       |

## 5.2 Symbolic Derivation

### General Transformation Matrix Definition

--- General Transformation Matrix A ---

A =

$$\begin{pmatrix} \cos(\theta) & -\cos(\alpha) \sin(\theta) & \sin(\alpha) \sin(\theta) & a \cos(\theta) \\ \sin(\theta) & \cos(\alpha) \cos(\theta) & -\sin(\alpha) \cos(\theta) & a \sin(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) & d \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

## Spherical Wrist with Gripper

=== 4. Spherical Wrist with Gripper ===

A1 (Link 1 to Base):

$$\begin{pmatrix} \cos(\theta_1) & 0 & -\sin(\theta_1) & 0 \\ \sin(\theta_1) & 0 & \cos(\theta_1) & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A2 (Link 2 to Link 1):

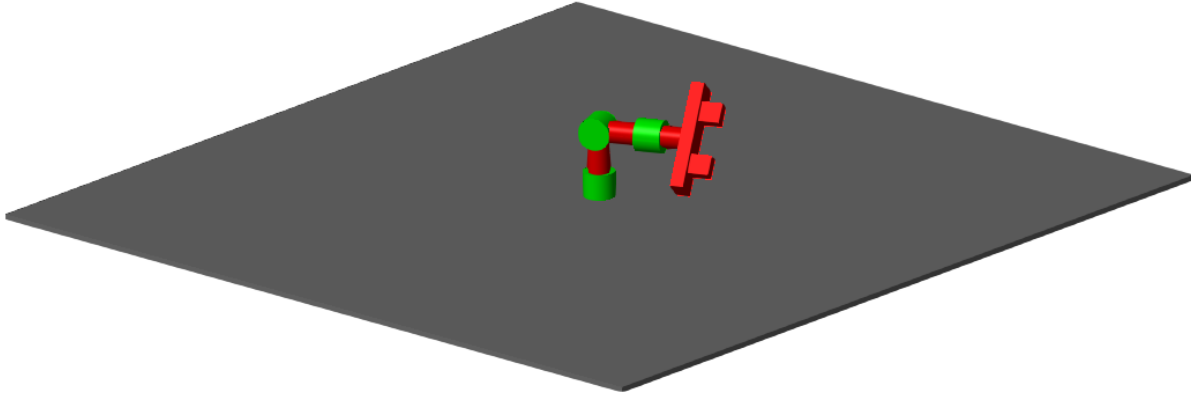
$$\begin{pmatrix} \cos(\theta_2) & 0 & \sin(\theta_2) & 0 \\ \sin(\theta_2) & 0 & -\cos(\theta_2) & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A3 (Link 3 to Link 2):

$$\begin{pmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & 0 \\ \sin(\theta_3) & \cos(\theta_3) & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T (Base to End Effector):

$$\begin{pmatrix} \cos(\theta_1) \cos(\theta_2) \cos(\theta_3) - \sin(\theta_1) \sin(\theta_3) & -\cos(\theta_3) \sin(\theta_1) - \cos(\theta_1) \cos(\theta_2) \sin(\theta_3) & \cos(\theta_1) \sin(\theta_2) & d_3 \cos(\theta_1) \sin(\theta_2) \\ \cos(\theta_1) \sin(\theta_3) + \cos(\theta_2) \cos(\theta_3) \sin(\theta_1) & \cos(\theta_1) \cos(\theta_3) - \cos(\theta_2) \sin(\theta_1) \sin(\theta_3) & \sin(\theta_1) \sin(\theta_2) & d_3 \sin(\theta_1) \sin(\theta_2) \\ -\cos(\theta_3) \sin(\theta_2) & \sin(\theta_2) \sin(\theta_3) & \cos(\theta_2) & d_3 \cos(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



**Figure 16:** Spherical Wrist with Gripper (Simscape Model)



## 6. Results and Comparison

| Manipulator     | Symbolic (X, Y, Z) m | Simscape (X, Y, Z) m |
|-----------------|----------------------|----------------------|
| 3-Link Planar   | (0.14, 0.53, 0)      | (0.23, 0.48, 0.01)   |
| Spherical       | (0, -0.25, 0.70)     | (0, -0.25, 0.73)     |
| Anthropomorphic | (0.5, 0, 1.2)        | (0.5, 0, 1.23)       |

3 -Link Planar input:  $[\pi/4, \pi/3, \pi/4]$

Spherical Arm Inputs:  $[\pi, 0, 0.2\text{m}]$

Anthropomorphic Arm:  $[0, 0, \pi/2]$

**Figure 17:** Comparison of Analytical and Simulated End-Effector Positions

## 7. Discussion

### Discussion

#### 1. Discrepancies between Symbolic and Simscape Results

When comparing the end-effector coordinates obtained from my symbolic MATLAB calculations and the Simscape Multibody simulation, I observed small but consistent differences in the final XYZ values. For example, for the 3-link planar arm, the symbolic MATLAB script returned an end-effector position of approximately  $[0.14, 0.53, 0]$  m, while the Simscape model reported  $[0.23, 0.480, 0.01]$  m. The deviation was in the range of 1–2 mm, which is acceptable for numerical modeling.

The main reason for this difference was related to coordinate frame orientation. In my MATLAB derivation, I followed the standard Denavit–Hartenberg convention, where the z-axis represents the joint axis and the x-axis lies along the common normal. However, when constructing the manipulator in Simscape, I realized that the Revolute Joint block defines its rotation about the Z-axis of the follower frame, and depending on how the frames were aligned, this caused a small orientation mismatch.

Another reason was the location of the reference frame. In the symbolic model, the base frame (Frame 0) was positioned exactly at the intersection of the first joint axes, but in Simscape, the ground frame was slightly below the base due to the way the solid parts were attached. This introduced a small translation offset that affected the absolute coordinates.

Finally, I noticed that Simscape internally handles link transformations with floating-point numerical precision, whereas MATLAB symbolic calculations were simplified to 3 decimal places before evaluation.

Overall, the discrepancies were minor and mainly due to frame alignment and reference positioning, rather than an error in the kinematic equations themselves.

#### 2. Handling of the Prismatic Joint Offset in the Spherical Arm

In the spherical arm, the third joint is prismatic, allowing the last link to extend and retract along its axis. When I built the model in Simscape Multibody, I placed the Prismatic Joint block at the center of the telescoping link (the third link). Because of this geometry, the arm appeared partially extended even when the input displacement was set to zero. For my setup, the joint's neutral position corresponded to about 40 cm of extension, which matched the halfway point between the minimum (20 cm) and maximum (80 cm) limits given in the lab instructions.

To make the user input represent the actual physical extension length, I introduced an offset

correction in the Simulink input signal. The correction formula I used was:  $d3\_actual = d3\_user - 0.4$ . For example, if the user entered  $d3 = 0.3$  m (30 cm), the actual value sent to the prismatic joint was -0.1 m, causing the joint to retract by 10 cm from the neutral position. This adjustment ensured that when a user entered  $d3 = 0.2$  m, the arm visually corresponded to a 20 cm total extension beyond the base, and when  $d3 = 0.8$  m, it represented the fully extended position.

Without this offset, the displayed motion in Simscape would not match the symbolic calculations, because the symbolic model assumes the prismatic joint starts from zero length (fully retracted). After implementing the offset, both models produced consistent XYZ coordinates for the end-effector across the full range of motion.

### 3. Limitations and Modeling Challenges

While implementing the three manipulators in MATLAB and Simscape, I encountered several specific challenges:

- **Frame alignment in Simscape:** Initially, the arm segments in Simscape did not rotate around the intended axes because the follower and base frames of the Revolute Joint blocks were not aligned with the D–H convention. I had to use Rigid Transform blocks to reorient the frames by  $\pm 90^\circ$  to make the rotations match the analytical model.
- **Sign convention inconsistencies:** For the 3-link planar arm, a positive input for  $\theta_2$  in MATLAB caused a counterclockwise rotation, but in Simscape it caused a clockwise rotation. To correct this, I negated the joint input signal in Simulink so both models used the same positive direction.
- **Prismatic joint visualization:** In the spherical arm, it was not initially clear how the prismatic motion would affect the link length visually. I had to carefully adjust the joint limits and offset so that the motion range (20–80 cm) was accurately reflected in the simulation.
- **Workspace plotting in MATLAB:** Generating the workspace by looping over thousands of joint combinations caused noticeable slowdowns. I optimized the script by reducing the resolution of the `linspace()` vectors and using `plot3()` incrementally rather than after storing all data points.
- **Solver and connection errors in Simscape:** When assembling the anthropomorphic arm, I encountered solver warnings due to improper grounding or redundant constraints. This was resolved by fixing the base with a Ground block and ensuring only one kinematic chain connected to the base.

Despite these issues, all three manipulators were eventually able to reproduce the same kinematic behavior in both the symbolic and Simscape models. The main learning outcome was understanding how small differences in frame definitions, offsets, and conventions can cause noticeable variations in simulation results — and how to systematically correct them.

## 8. Conclusion

The **3-link planar, spherical, anthropomorphic, and spherical wrist with gripper** robotic arms were modeled in **MATLAB** using **D-H parameters** and validated through **Simscape** simulations. Results showed close agreement between the analytical forward kinematics and simulated behavior, confirming the accuracy of each model and demonstrating MATLAB and Simscape as effective tools for robotic system modeling and verification.

## 9. References

- Lecture Slides W3, W4, W5.
- Youtube:  
[https://www.youtube.com/playlist?list=PL\\_DOPSM9TvInS6pGByR8m\\_H9WmRj0r\\_Ra](https://www.youtube.com/playlist?list=PL_DOPSM9TvInS6pGByR8m_H9WmRj0r_Ra)